

ANALYSTLAB AFRICA

Experience Lab Internship Programme

Week 5: Exploratory Analysis, KPI Development & Business Insights Data Analytics Track — Initial HealthConnect Analytics Report

Field	Detail
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Prepared by	Oluebube
Track	Data Analytics
Week	Week 5

1. Data Preparation Summary

Building on the Week 4 initial data quality assessment, the HealthConnect Appointment Dataset was reviewed and prepared for exploratory analysis and KPI calculation. The original dataset was not overwritten; all preparation and calculation steps were carried out on working copies of the data.

Key data preparation observations carried forward from Week 4 include:

- 5,000 appointment records across 18 variables, with 1,696 unique patients and 5,000 unique appointment IDs — no duplicate records were found.
- Missing values were limited to Distance to Clinic (90 records, 1.80%) and Waiting Time (60 records, 1.20%); all other variables were complete.
- Date fields (booking_date, appointment_date) and derived fields (appointment_day, booking_lead_days) were confirmed internally consistent, with 0 mismatches found in each check.
- Categorical fields (gender, appointment_type, appointment_time, reminder_sent, reminder_channel, appointment_outcome) contained clean, consistent category values with no spelling variants.

For Week 5, the dataset was grouped and aggregated to support KPI calculation and factor investigation — for example, grouping previous_appointments and previous_no_shows into bands, and grouping booking_lead_days and distance_to_clinic_km into ranges — using Power Query and Power BI. These groupings were applied without altering the original source file.

2. Exploratory Data Analysis

The exploratory analysis investigated appointment attendance and no-show patterns across time, patient characteristics, previous appointment history, reminders, booking behaviour, appointment scheduling, and accessibility factors.

2.1 Monthly Appointment Volume and No-Show Trend

Monthly appointment volume declined over the course of the year, from a high of 585 appointments in June to a low of 260 appointments in December. The monthly no-show rate fluctuated throughout the year without a steady upward or downward trend, but reached its highest point in December at 57.61%.

2.2 Previous No-Show Behaviour

Patients with a history of previous missed appointments (up to 5 previous no-shows) showed the highest no-show rates, while patients with no previous missed appointments showed the lowest no-show rates. This suggests a strong association between past no-show behaviour and the likelihood of a current appointment being missed.

2.3 Age Group and Gender

The 55–64 age group recorded the highest no-show rate at 53%, closely followed by the 25–34 age group at 52.93%. However, no-show rates across all age groups were closely clustered, with no clear increasing or decreasing pattern by age.

By gender, males recorded the highest no-show rate (51.50%), followed by females (51.10%). Patients who preferred not to disclose their gender had the lowest no-show rate of the three categories.

2.4 Reminders and Reminder Channel

Patients who received a reminder had a lower no-show rate (49.89%) compared with those who did not receive a reminder (54.63%), suggesting reminders are associated with improved attendance.

Among reminder channels, SMS was associated with the lowest no-show rate, WhatsApp with the highest, and Email fell in between.

2.5 Booking Lead Time

As booking lead time increased, the no-show rate generally increased as well, suggesting that appointments booked further in advance are more likely to be missed than those booked closer to the appointment date.

2.6 Appointment Day

No-show rates were highest on Monday, Sunday, and Wednesday (53.13%, 52.84%, and 52.81% respectively), and lowest on Friday (48.71%).

2.7 Distance to Clinic

As distance to the clinic increased, the no-show rate generally increased — appointments within 30–40km recorded a no-show rate of 73.77%, compared with 49.24% for appointments below 10km. However, the 40–50km distance band did not follow this trend, recording a lower no-show rate of 44.4%, which stands out as an anomaly worth further investigation.

2.8 Waiting Time

No consistent relationship was found between waiting time and no-show rate — the rate neither increased nor decreased steadily with waiting time. Appointments with a waiting time below 10 minutes had the highest no-show rate (52.71%), while those with a waiting time between 50–60 minutes had the lowest (49.42%).

2.9 Appointment Type

No-show rates were highest for Follow-up appointments and lowest for General Consultation appointments.

3. KPI Calculations

The following five KPIs were selected from the potential KPIs proposed in Week 4, based on their relevance to the appointment attendance business questions. Each KPI is defined, calculated, and interpreted below.

3.1 Total Appointments

Definition:

The total number of appointments recorded in the dataset.

Why it matters:

Establishes the overall scale of appointment activity and provides the base for all other KPI calculations.

Calculated Result:

5,000

Business Question:

Overall appointment volume.

Interpretation:

The dataset covers a substantial volume of appointments, providing a reliable base for calculating attendance-related rates.

3.2 Attendance Rate (%)

Definition:

The proportion of non-cancelled appointments that were attended.

Why it matters:

Directly measures how often patients honour their scheduled appointments.

Calculated Result:

48.85%

Business Question:

Are patients attending their appointments?

Interpretation:

Fewer than half of non-cancelled appointments were attended, indicating that attendance is a significant operational concern for HealthConnect.

3.3 No-Show Rate (%)

Definition:

The proportion of non-cancelled appointments that were missed.

Why it matters:

Directly measures the scale of the no-show problem, which is the central concern of this project.

Calculated Result:

51.15%

Business Question:

How serious is the no-show problem?

Interpretation:

More than half of all non-cancelled appointments were missed, confirming that no-shows are a major and pressing operational challenge for HealthConnect.

3.4 Cancellation Rate (%)

Definition:

The proportion of all appointments that were cancelled.

Why it matters:

Distinguishes planned cancellations from unplanned no-shows, ensuring the no-show rate reflects genuine missed appointments rather than appointments that were formally cancelled in advance.

Calculated Result:

5.26%

Business Question:

How common are cancellations?

Interpretation:

Cancellations make up a relatively small share of all appointments, confirming that the majority of the attendance problem stems from no-shows rather than cancellations.

3.5 Previous No-Show Rate (%)

Definition:

The proportion of patients with at least one previous missed appointment.

Why it matters:

Indicates how common repeat no-show behaviour is among patients, and supports the investigation of whether past behaviour predicts future attendance.

Calculated Result:

41.58%

Business Question:

Are previous no-shows an important behaviour to investigate?

Interpretation:

A substantial share of patients have a history of missed appointments, and this group is associated with notably higher current no-show rates — making previous no-show history a potentially valuable factor for identifying at-risk patients.

4. Visualisations / Dashboard

An initial analytical dashboard was developed in Power BI to bring together the KPIs and exploratory findings above. The dashboard is in the pdf file : [dashboard.pdf](#)

5. Key Findings

- Nearly half of all non-cancelled appointments (51.15%) end in a no-show, confirming appointment attendance as a major operational issue for HealthConnect.
- Previous no-show history is strongly associated with current attendance behaviour: patients with a history of missed appointments show markedly higher no-show rates than patients with no such history, and 41.58% of patients fall into this at-risk group.
- Reminders are associated with meaningfully better attendance: patients who received a reminder had a no-show rate almost 5 percentage points lower than those who did not (49.89% vs. 54.63%), and SMS reminders performed best while WhatsApp reminders performed worst.
- Longer booking lead times are associated with higher no-show rates, suggesting that appointments booked far in advance carry greater attendance risk.
- No-show rates vary meaningfully by day of the week (highest on Monday, Sunday, and Wednesday; lowest on Friday) and by appointment type (highest for Follow-up, lowest for General Consultation).
- Distance to the clinic is broadly associated with higher no-show rates, though the 40–50km band breaks this pattern and warrants further investigation rather than being treated as a reliable trend.
- Demographic factors (age group, gender) show only mild variation in no-show rate, and waiting time shows no consistent relationship with no-show behaviour — neither appears to be a strong standalone driver of attendance.

6. Business Recommendations

- Prioritise reminder outreach — and specifically SMS reminders — for all patients, since reminders are associated with a clear reduction in no-show rate and SMS outperforms other channels.
- Reassess the use of WhatsApp as a reminder channel, or investigate why it is associated with the highest no-show rate, before scaling it further as a primary communication method.
- Introduce a patient risk-flagging approach based on previous no-show history, allowing staff to prioritise additional confirmation calls or reminders for patients with a history of missed appointments.

- Review scheduling and reminder timing for appointments booked with long lead times, since longer lead times are associated with higher no-show risk.
- Consider adjusted staffing or overbooking strategies for Monday, Sunday, and Wednesday appointments, where no-show rates are highest, and for Follow-up appointments specifically.
- Investigate accessibility support (e.g. transport assistance or telehealth options) for patients travelling longer distances to the clinic, while further examining the anomaly in the 40–50km distance band before drawing firm conclusions.

7. Cross-Track Collaboration

HealthConnect is a shared, multidisciplinary project. As part of Week 5, the Data Analytics track identified a meaningful collaboration point with the Data Science track.

1. Track Interacted With:

Data Science Track.

2. Information/Output Exchanged:

The Data Analytics track shared its KPI results and key exploratory findings with the Data Science track — specifically, the variables found to have the strongest observed association with appointment no-shows: previous no-show history, reminder status and reminder channel, booking lead time, appointment day, and appointment type. The anomaly identified in the 40–50km distance band, and the lack of a consistent relationship between waiting time and no-show rate, were also flagged.

3. Why the Interaction Was Relevant:

The Data Science track's Week 4 output required proposing a target variable and potential input features for a no-show prediction model. The Data Analytics track's exploratory findings provide empirical, data-driven evidence of which variables show a genuine relationship with the outcome variable (appointment_outcome), giving the Data Science track a grounded starting point for feature selection rather than relying on assumptions alone.

4. What Changed or Improved as a Result:

The exchange helps the Data Science track prioritise previous_no_shows, reminder_sent, reminder_channel, booking_lead_days, appointment_day, and appointment_type as likely high-value input features for the prediction model. It also flags distance_to_clinic_km and waiting_time_minutes as variables that may require closer inspection or additional feature engineering, since their relationship with no-show behaviour was inconsistent or anomalous in the exploratory analysis rather than straightforwardly linear.

8. Review of Assumptions, Limitations, Risks and Dependencies

The assumptions, limitations, risks, and dependencies identified in Week 4 were reviewed in light of the Week 5 exploratory analysis and KPI results. Each item is marked as Still Relevant, Resolved, or Changed, with notes on impact and how it should be addressed going forward. New issues discovered during Week 5 are documented separately at the end of this section.

8.1 Assumptions — Review

Assumption (Week 4)	Status	Impact / How to Address
Appointment outcome categories are correctly recorded in the dataset.	Still Relevant	Remains an unverifiable assumption, as no independent source exists to confirm outcome accuracy. It should continue to be stated as an assumption in all future reporting.
Each appointment record represents a unique scheduled appointment.	Resolved	Confirmed by the Week 4 duplicate check (0 duplicate records, 0 duplicate appointment IDs). No further action needed.

Assumption (Week 4)	Status	Impact / How to Address
Variables in the dataset are defined consistently with the HealthConnect Data Dictionary.	Changed	Largely holds, but the exact definition of waiting_time_minutes (e.g. whether it captures time before or after arrival) was never fully confirmed against the Data Dictionary. This should be verified before waiting time is used more heavily in later analysis.
The available appointment and patient information is sufficient to investigate potential no-show patterns.	Resolved	Confirmed by Week 5: multiple meaningful associations were identified (previous no-shows, reminders, booking lead time, appointment day, distance), demonstrating the dataset was sufficient for the business questions investigated.

8.2 Limitations — Review

Limitation (Week 4)	Status	Impact / How to Address
The analysis is based only on the variables available in the HealthConnect dataset.	Still Relevant	Unchanged. Factors such as patient health status, transportation access, or work schedules remain uncaptured and should be noted as a scope boundary in the final report.
The analysis identifies associations and patterns rather than proving causation.	Still Relevant	Reinforced by Week 5 findings — for example, the reminder and booking-lead-time relationships are associations, not proven causal effects. All business recommendations should continue to be framed accordingly.
Some variables may contain missing or incomplete information.	Still Relevant	Confirmed missingness in Distance to Clinic (1.80%) and Waiting Time (1.20%) persisted into Week 5 and may partly explain the distance anomaly at 40–50km. Should be explicitly noted wherever these two variables are used.
The analysis is based on historical appointment data.	Still Relevant	Unchanged. Findings should be periodically re-validated against newer data as it becomes available, rather than treated as permanently fixed.

8.3 Risks — Review

Risk (Week 4)	Status	Impact / How to Address
Poor data quality	Resolved	Week 4's data quality checks found the dataset largely clean (0 duplicates, only minor missingness), lowering this risk. Continued monitoring is still advisable if the dataset is refreshed or extended.
Missing information	Still Relevant	Missing values in Distance to Clinic and Waiting Time remain. Mitigation continues to be explicit documentation of these gaps wherever the affected variables are analysed.
Incorrect interpretation	Changed	This risk materialised in Week 5: the 40–50km distance band did not follow the otherwise increasing no-show trend. It was flagged as an anomaly rather than smoothed into the overall pattern, which is the correct mitigation, but

Risk (Week 4)	Status	Impact / How to Address
		it should be investigated further before being used in any recommendation.
Inconsistent variable categories	Resolved	Week 4's categorical checks found no spelling variants or inconsistent categories across gender, appointment type, reminder status, or outcome fields. No further action needed.
Misclassification of appointment outcomes	Still Relevant	Category consistency was confirmed, but the underlying accuracy of outcome recording (e.g. whether a 'No-Show' was correctly logged) cannot be independently verified and remains a low-level risk.

8.4 Dependencies — Review

Dependency (Week 4)	Status	Impact / How to Address
Availability and accessibility of the HealthConnect Appointment Dataset.	Resolved	The dataset was available and used throughout Week 5 without access issues.
Accurate variable definitions in the HealthConnect Data Dictionary.	Still Relevant	The waiting_time_minutes definition remains unconfirmed against the Data Dictionary. This should be resolved before waiting time is weighted more heavily in future analysis or modelling.
Appropriate analytical tools and software.	Resolved	Excel and Power BI/Power Query were used successfully to complete the Week 5 data preparation, KPI calculations, and dashboard.
Sufficient data quality for meaningful analysis.	Resolved	Confirmed by the Week 4 data quality assessment and the successful completion of Week 5 EDA and KPI calculations.
Accurate recording of appointment outcomes and relevant patient information.	Still Relevant	Remains an unverifiable assumption/dependency, as with 9.1 above; continues to be stated as a standing dependency of the analysis.
Availability of the variables required to investigate the identified business questions and calculate the selected KPIs.	Resolved	All five proposed KPIs were successfully calculated, and all primary business questions from Week 4 were investigated using the available variables.

8.5 New Issues Discovered in Week 5

- The 40–50km distance band does not follow the otherwise increasing relationship between distance and no-show rate (44.4% vs. 73.77% for 30–40km). This may reflect a small sample size in that band, a data quality issue, or a genuine but more complex relationship, and requires further investigation before being used in any recommendation.
- No consistent relationship was found between waiting time and no-show rate, which is a weaker signal than may have been anticipated in Week 4 when waiting time was proposed as a potential patient-experience factor. Its priority as an explanatory variable should be lowered accordingly.
- WhatsApp reminders are associated with the highest no-show rate among reminder channels, which may reflect the channel itself, patient demographics using that channel, or delivery reliability — this distinction has not yet been investigated and should be treated with caution rather than as a direct causal signal.
- The exact peak no-show rate associated with the longest booking lead times was not fully captured during this round of analysis and should be confirmed in a follow-up check.

9. Limitations

- The analysis identifies associations and patterns rather than proving causation; no factor discussed above should be interpreted as directly causing no-show behaviour.
- Missing values in Distance to Clinic (1.80%) and Waiting Time (1.20%) may have slightly affected the completeness of analyses involving these variables.
- The anomaly observed in the 40–50km distance band is not explained by the current analysis and may reflect a data quality issue, a small sample size in that band, or a genuine but more complex relationship that requires further investigation.
- The analysis is based on historical appointment data and may not fully reflect future patient behaviour or changes in HealthConnect's operations.
- The analysis is limited to the variables available in the HealthConnect dataset; other factors not captured in the data (for example, patient health status or transportation access) may also influence attendance.